ISSSP 2026 — TRACK 3 FINALS: ITEMS 5, 6 & 7 (WORK DISPLAY, PPT & ARTISTRY — 40 POINTS)

Presentation Script, Work Display & Simulation Guide

Complete 14-Slide Deck Blueprint, Word-for-Word 10-Minute Presentation Script, Judge Q&A; Defense, Live HUD/3D Demonstration Walkthrough, and Artistry Aesthetics Guide

Project: L-MINER Mk-II Autonomous Prospecting Payload	**Institution:** Institute of Space Technology (IST), Islamabad
Scoring Focus: Items 5 (25 Pts), 6 (10 Pts), 7 (5 Pts)	**Duration:** 10 Min Pitch + 5 Min Q&A; Defense

Part I: Work Display & Simulation Architecture (25 Points)

Table 6 awards 25 points for 'Work display / simulation design'. L-MINER demonstrates a dual-interface cyber-physical environment:
1. Interactive Mission Control HUD (dashboard.py): Real-time aerospace telemetry cards, dual GPR B-scan + A-scan energy traces, USGS laboratory benchmark overlay curves, autonomy state indicators, and survey sequence buttons.
2. High-Fidelity 3D WebGL Digital Twin (visualization_3d.py): 96x96 crater terrain mesh with 3° polar solar illumination, 6-wheel rocker-bogie rover mesh, terrain-following slope adaptation, and subterranean deposit ellipsoids sized by ML confidence.

Step	Simulation Action	Demonstration Focus for Judges
1. Boot	Launch HUD (:8501)	Avionics nominal (28°C), 100% battery, instruments standby.
2. Scan	Click '■ Scan Next'	Cell (0,1): GPR B-scan yellow horizon at 1.2 m, spectrometer ice dips at 1.25/1.5/2.0 µm, ML classifies Water Ice (98.4%).
3. Hazard	Encounter (0,3)	Obstacle score 0.80 >= 0.75 triggers L0 halt. Mobility safely protected.
4. Power	Battery drops <30%	L1 low-power hold engaged. Deploy solar panels for +25% recharge.
5. 3D Twin	Toggle 3D View	Rotate 3D scene: 3° polar shadows, rocker-bogie tilt, subterranean spheres.
6. Auto	Click '■ Auto-Survey'	Autonomous completion of 36 cells. Final summary: ~57.8 Tonnes Water Ice.

Part II: PPT Presentation Deck Structure & Speaker Script (10 Points)

Structured as a 14-slide presentation deck timed for exactly 10 minutes (plus 5 minutes for judge Q&A;). Speaking roles rotate among all four team members:

Slide & Timing	Speaker	Core Message & Technical Content
1. Title & Vision (0:00-0:45)	Haider	Introduction to L-MINER Mk-II: Autonomous lunar polar volatile prospecting payload.
2. Cold Traps (0:45-1:30)	Haider	The Ground-Truth Gap: Why orbital sensors cannot resolve subsurface ice volume.
3. System Concept (1:30-2:15)	Haider	Dual-modality payload: 500 MHz GPR (depth) + VNIR-SWIR Spectrometer (chemistry).
4. 3-Tier Autonomy (2:15-3:00)	Saad	L0 Safety Reflex (10 Hz), L1 Resource Manager (1 Hz), L2 Cognitive Science Layer.
5. Radar Physics (3:00-3:45)	Haider	500 MHz GPR wave propagation, dielectric contrast (Γ=0.85 for ice), B-scans.
6. Spectroscopy (3:45-4:30)	Haider	0.5–2.5 µm active spectrometry, USGS splib07 validation, diagnostic ice bands.
7. 10-D Features (4:30-5:15)	Wasay	3,300x edge compression: 32,968 raw floats reduced to 10 scalar features.
8. ML Engine (5:15-6:00)	Wasay	Random Forest (100 trees), 95.8% accuracy, sub-15 ms inference latency.
9. LIVE DEMO (6:00-7:30)	Hajra/Saad	Live demonstration of Mission Control HUD and 3D WebGL Digital Twin.
10. ISRU Yield (7:30-8:15)	Hajra	~57.8 Tonnes Water Ice localized; yields 51.4 T LOX + 6.4 T LH2 propellant.
11. Benchmarking (8:15-8:45)	Wasay	14x faster throughput vs. teleoperated rovers, zero latency bottlenecks.
12. Team & Agile (8:45-9:15)	Hajra	Interdisciplinary collaboration at IST Islamabad, daily standups, Git PRs.
13. Roadmap (9:15-9:45)	Haider	TRL 4 -> 6: Cryo-vacuum simulant bed testing, NASA Core Flight System port.
14. Conclusion (9:45-10:00)	Haider	Final mission summary & transition to 5-minute Judge Q&A; Defense.

Anticipated Judge Q&A; Defense

Judge Question	Team Defense Answer
Q1: Void vs. Ice in GPR?	**Haider:** A void creates a dielectric transition to vacuum (ε_r=1.0) with phase inversion. Ice permafrost (ε_r=3.1) exhibits a forward phase step. Crucially, the spectrometer concurrently tests for molecular O-H absorption at 1.5/2.0 µm, eliminating false positives.
Q2: Recharging in PSR darkness?	**Saad:** Opportunistic crater-corridor traverse. The rover dips into cold traps for science sounding on battery power. Before SoC drops <30%, L1 commands ascent back to crater rim 'Peaks of Eternal Light' to deploy solar arrays.
Q3: Why Random Forest over CNN?	**Wasay:** Spaceflight CPUs have strict <10W power budgets. Our 10-D feature reduction runs Random Forest in <15 ms, resists overfitting on sparse extraterrestrial training data, and provides explainable decision trees essential for flight qualification.

Part III: Artistry, Aesthetics & Visual Identity (5 Points)

1. Mission Control Dark Theme: Deep space background (#0C1527 to #010409) creates an authoritative aerospace command center look, minimizing eye fatigue during dark-room judging.
2. Functional Semantic Colors: Cyan (#00E5FF) for water ice and GPR radar; Emerald (#00E676) for nominal status; Amber (#FF9100) for ilmenite and power alerts; Crimson (#FF1744) for obstacle hazards.
3. Technical Typography: 'Space Grotesk' for bold mission titles paired with 'JetBrains Mono' for telemetry readouts.
4. 3D Visual Polish: Rocker-bogie mesh with gold MLI foil, silicon panels, 3° polar shadows, and confidence-scaled subsurface deposit ellipsoids.